User Guide

A plain-language walkthrough of what the tool shows, how to use it, and how to read the maps responsibly.

The Heat Risk Explorer is an interactive map that helps you see which San Diego neighborhoods face the most danger during extreme heat — and why.

Pick the things you care about — temperature, AC access, tree cover, age, chronic illness, income, and more — and the tool builds a neighborhood-level risk map you can read at a glance.

You don't need to know statistics or mapping software to use it. Everything runs in your web browser. There's nothing to install and nothing to download.

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Try it in 60 seconds

Want to skip ahead? Here's the fastest way to see what the tool does:

1. **Open the map** at connorjmack.github.io/heat-risk-viewer.
2. **Leave the default boxes checked.** They're a reasonable starting place.
3. **Click "Build Composite Risk Map."** It's the big blue button on the left.
4. **Look for the dark red areas.** Those are the neighborhoods with the highest overall risk.
5. **Click any neighborhood.** A small box pops up with its scores and the biggest reason it scored that way.

That’s it. The rest of this guide explains what you’re looking at and how to dig deeper.

Why heat hits some neighborhoods harder

A 100° day is not the same everywhere. A shaded apartment in Carlsbad with working AC and a healthy adult inside is one story. A second-floor unit in El Cajon with no AC, a 90-year-old resident, and a hot roof above is another story altogether.

To make sense of that, the tool breaks heat risk into three pieces:

- **How hot it gets** — the actual heat the neighborhood experiences.
- **Who lives there** — how many people are likely to be harmed when temperatures spike.
- **What helps people cope** — the things in a neighborhood that protect people, like AC, tree shade, and nearby healthcare.

Every neighborhood gets a score on each of these three pieces. The tool then blends them into one overall heat risk score from 0 (lowest risk) to 100 (highest risk).

What goes into each piece

> ### How hot it gets shown in red
>
> This is the physical heat a neighborhood experiences. We measure it using weather stations across the county:
>
> - **Daytime highs** — average summer highs, the hottest days, and the single hottest day of the year. *Best when you’re focused on peak heat events or short-term emergency planning, like deciding when to open cooling centers.*
> - **Nighttime lows** — how cool nights actually get. Bodies need cool nights to recover. When nights stay warm, heat illness goes up — especially for older adults. *Best when you’re worried about multi-day heat waves or older-adult health, where the lack of overnight relief is what makes people sick.*
> - **Humidity** — how heavy and muggy the air is. High humidity makes the same temperature feel worse. *Best when you’re comparing inland and coastal neighborhoods, where the same air temperature can feel very different.*

Who lives there shown in orange

This is about the people in a neighborhood. Some are more likely to be harmed by heat than others. The tool looks at:

- **Age** — kids under 18 and adults over 65 are more vulnerable. *Best when you're trying to see where heat illness is most likely to land — senior housing, family neighborhoods, areas near schools.*
- **Health conditions** — heart disease, asthma, COPD, diabetes, high blood pressure, and others get worse in heat. *Best when you're targeting healthcare partnerships, medication-safety messaging, or hospital-readiness planning.*
- **Daily life challenges** — language barriers, disability, poverty, no car, housing cost stress, and other things that make hot days harder to navigate. *Best when you're planning heat communications, transit to cooling centers, or community navigator programs.*
- **Air quality and life expectancy** — neighborhoods already living with worse air or shorter life expectancy tend to be hit harder. *Best when you're aligning heat work with broader environmental justice or health equity goals.*

What helps people cope shown in teal

This is the protective side of the equation — what a neighborhood has going for it:

- **AC at home** — how likely a home is to have any AC, central AC, or none at all. *Best when you're prioritizing cooling-center locations, utility bill assistance, or home AC subsidy programs.*
- **The neighborhood itself** — tree cover, parks, green space, and the age of the housing (older homes are usually less insulated). *Best when you're planning tree planting, park investment, cool-roof programs, or weatherization.*
- **Money and services** — income, education, health insurance, healthcare access, food access, and how connected the community is. *Best when you want to braid heat work with other safety-net programs, or see where lack of resources amplifies the impact of a heat wave.*

More of these means lower heat risk. The tool flips the math automatically, so you don't have to think about it.

How the tool puts it all together

Once you've picked your factors, the tool blends them into one easy-to-compare score for every neighborhood — from 0 (lowest risk) to 100 (highest risk).

Two things make this smarter than a simple average:

It handles overlap. Many of these factors travel together. Lower-income neighborhoods often also have older housing, less tree cover, and higher rates of chronic illness. If we just added all those numbers up, we'd be counting the same underlying pattern several times. The tool finds those overlapping patterns and counts them once.

It gives credit where it's due. Things like AC, tree cover, and nearby healthcare protect people. The tool subtracts those from the risk total, so a neighborhood with strong protections ends up with a lower score even if it gets hot.

The result is one number per neighborhood — easy to see, click, and compare across the county.

> **For the curious:** the technique behind this is called Principal Component Analysis (PCA). You don't need to know how it works to use the tool, but if you'd like a friendly visual explainer, this one from Setosa is a good starting place.

Reading the map

The colors

Darker always means higher. On the overall composite map, dark red means a neighborhood is among the highest risk in the county. On the individual pillar maps, the color follows the pillar — red for heat, orange for population vulnerability, teal for protective resources.

What a score actually means

The scores are *relative*, not absolute. A neighborhood with a score of 75 is at higher risk than 75% of other neighborhoods in San Diego County. It does **not** mean 75% of residents will be harmed.

Clicking a neighborhood

Click any neighborhood and a small box appears with:

- Its score on each of the three pieces
- Its overall composite score
- Its **dominant driver** — the piece pushing its risk the highest. This answers the question, "Why is this neighborhood in the red?"

The map layers

- **Composite Risk** — the overall 0–100 score that combines everything. The default.
- **Dominant Driver** — colors each neighborhood by which of the three pieces matters most there. Helps you see whether a hotspot is hot because of physical heat, who lives there, or a lack of resources.
- **Individual pillar views** — show one piece (heat, who lives there, or what helps them cope) on its own.

Looking into the future

At the top of the left panel, four buttons let you switch between time periods:

- **Present Day** — what we measure today, using actual weather station readings.
- **Moderate Emissions** — a middle-of-the-road future where the world cuts emissions but not aggressively.
- **High Emissions** — a future where emissions stay high.
- **Very High Emissions** — a worst-case future of unchecked fossil fuel use.

When you pick a future scenario, the tool shifts the temperature numbers using climate model projections for either 2050 or 2080. Everything else — who lives where, what resources are around — stays at today's values. We don't pretend to know who will live in each neighborhood thirty years from now.

Three things you can do with this

Find a good spot for the next cooling center

1. Start with the default selections.
2. Under **What helps people cope**, uncheck the AC variables. You want a view that penalizes the *lack* of AC instead of treating homes that have AC the same as ones that don't.
3. Click **Build Composite Risk Map**.
4. Look at the darkest red neighborhoods.
5. Click a few. If their dominant driver is the "what helps people cope" piece, they're strong cooling-center candidates. If it's something else, a cooling center alone won't fully solve the problem.

See which neighborhoods will get hottest by mid-century

1. Click **High Emissions** at the top of the left panel.
2. Uncheck everything except the temperature variables.
3. Click **Build Composite Risk Map**.
4. Open the tool in a second browser tab on **Present Day** to compare side-by-side. The difference shows you where warming is concentrated.

Make the case for tree planting in one neighborhood

1. Use the default selections and click **Build Composite Risk Map**.
2. Click the neighborhood you care about.
3. Note its overall score, its dominant driver, and how its tree cover compares to nearby neighborhoods.
4. Take a screenshot. The map plus this guide is your documentation.

What this tool is — and isn't

It is:

- A way to see patterns across San Diego at a neighborhood scale.
- A starting point for conversations about where to focus heat preparation.
- A tool that lets you decide which factors matter to your question.

It isn't:

- A real-time heat warning. For active heat events, follow the National Weather Service.
- A guarantee about any one person's risk. A neighborhood score can hide the single block that's most exposed.
- A replacement for talking to communities. The neighborhoods showing the highest risk are the ones whose voices should shape what's done about it.

Common questions

Why are some neighborhoods blank on the map?

A neighborhood shows up blank when one of the variables you've selected is missing for it. Try unchecking the variable with the spottiest coverage and the blanks should fill in.

Can I save my map?

Not yet. For now, take a screenshot. The tool runs entirely in your browser, so there's no account to log into and nothing saved on a server.

Can I use this for a different county?

Not as-is. Every variable comes from a national dataset, but someone would need to assemble the data for a different county. The method is portable; the prepared data isn't.

What's the difference between "Composite Risk" and "Dominant Driver"?

Composite Risk shows the overall score — how hot, how vulnerable, and how protected, all combined. Dominant Driver shows *which of those three pieces* is the biggest in each neighborhood. Composite tells you *where* to focus; Dominant Driver tells you *what kind of solution* might help.

Who do I contact about this tool?

Feedback, bug reports, and suggestions are welcome via GitHub issues.

Where the data comes from

Every number in this tool comes from a public, trustworthy source.

WHAT IT MEASURES	SOURCE
Weather observations	MesoWest / Synoptic Data weather station network
Future temperature projections	LOCA2 downscaled climate model output (CMIP6)
Land surface temperature	USGS Landsat satellite imagery
Chronic disease rates	CDC PLACES
Social vulnerability	CDC Social Vulnerability Index (SVI) 2022
Income, education, healthcare, housing	Healthy Places Index (HPI)
Demographics and housing	American Community Survey (5-year estimates)
Air conditioning prevalence	Modeled estimates from Romitti et al. and Ahn et al.

Each variable in the tool also has a small `src` link next to its name that opens the original source in a new tab — so you can always trace a number back to where it came from.

A few words you might see

Census tract

A small Census Bureau geography, usually around 4,000 people. San Diego County has roughly 630 of them.

Composite risk score

The tool's final 0–100 score combining all three pieces.

Dominant driver

The piece (heat, who lives there, or what helps them cope) that contributes the most to a neighborhood's score.

Percentile

A score from 0 to 100 telling you where a neighborhood sits compared to all others in the county. A score of 80 means higher than 80% of other neighborhoods.

Climate scenario / SSP

A "what if" path for future emissions used by climate scientists. Higher SSP numbers mean more warming.

Principal Component Analysis (PCA)

The math the tool uses to find the patterns hidden in many overlapping variables. You don't have to understand it to use the tool. (visual explainer)